

Tractable Problems in AI Security via Formal Methods

A sane way to burn a few postdoc-years

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Quinn Dougherty

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1 The strategic landscape

Let us treat ML training and inference with the seriousness of airplanes



Authors

Quinn Dougherty, Max von Hippel, Gregory Malecha, Nora Ammann

We mostly write about model weight confidentiality and integrity, but other notions of AI security are in scope. Not the biggest possible tent, but more than two invariants.



↶ PRESIDENTIAL ACTIONS

PROMOTING ADVANCED ARTIFICIAL INTELLIGENCE INNOVATION AND SECURITY

Executive Orders | June 2, 2026

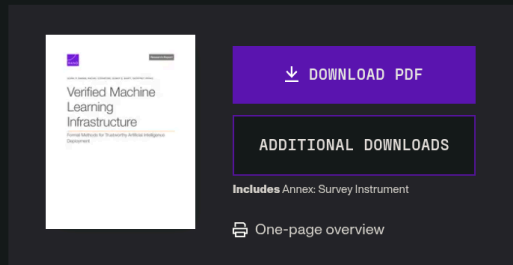
Figure 1: Demands hardened critical infrastructure, doesn't say how.

Verified Machine Learning Infrastructure

Formal Methods for Trustworthy Artificial Intelligence Deployment

[Gopal P. Sarma](#), [Rachel Steratore](#), [Sunny D. Bhatt](#), [Geoffrey Irving](#)

RESEARCH — Published Jun 4, 2026



The screenshot shows the RAND report interface. On the left is a thumbnail of the report cover, which features the title 'Verified Machine Learning Infrastructure' and the subtitle 'Formal Methods for Trustworthy Artificial Intelligence Deployment'. To the right of the thumbnail are two main interactive elements: a purple button with a download icon and the text 'DOWNLOAD PDF', and a white button with a purple border and the text 'ADDITIONAL DOWNLOADS'. Below the 'ADDITIONAL DOWNLOADS' button, there is a link that says 'Includes Annex: Survey Instrument'. At the bottom right of the section, there is a link with a document icon and the text 'One-page overview'.

Figure 2: Expert survey, human and simulated/silicon

ARIA

TODO



Anthropic's Frontier Safety Roadmap

We believe that AI capabilities will improve rapidly in the coming years. We will need to quickly and dramatically improve our state of preparedness in a number of areas, especially:

Security

Preventing theft, sabotage and/or manipulation of our AI models.

Safeguards

Preventing dangerous use of our models via product surfaces and within Anthropic itself.

Figure 3: See *Leveling up across the board* at <https://www.anthropic.com/responsible-scaling-policy/roadmap>

Speed

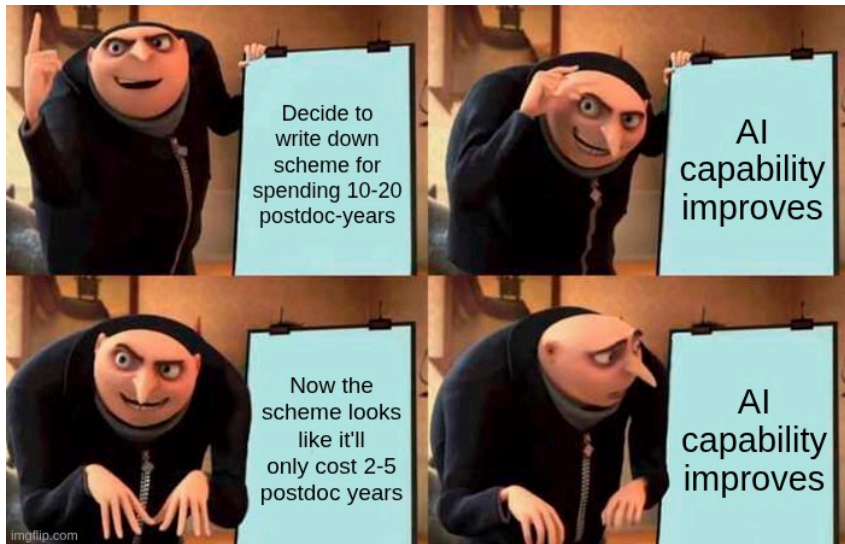


Figure 4: This is technically a violation of the meme format.

2 The stack

5 layers



Figure 5: The path of a prompt from the enduser to the chip's circuits

3 Tractable Problems

Hardening protocol boundaries



TODO

Hardening OCI runtime



TODO

Device drivers for verified kernels



The kernel can be verified; the driver underneath it usually can't.

Status quo

Multi-tenant GPU isolation rides on million-line hypervisor TCBs, with a proprietary ring-0 GPU driver as the single point of compromise.

The good news

seL4, NOVA, seKVM, and AWS Nitro verify a minimal core and push drivers out to deprivileged user-mode — the right architecture.

The real blocker

Not the proof, the spec. Driver proofs are a settled methodology (a verified ZynqMP DMA engine exists); what's missing is a machine-checkable model of the GPU's command / DMA / IOMMU surface. Vendors model the **ISA** — Arm, RISC-V, even AMD's shaders — not the device.

Device drivers for verified kernels (ii)



The widget

Specify one open GPU stack's command-submission ring (NVK/Nouveau) in Rocq/Lean; prove no guest command can drive a DMA or IOMMU mapping outside its region. Stub the hardware model first (reference), then co-develop a real one (research).

4 Onwards

How you can help



More contributions

Come aboard! More room for authors. Or, if you want to be low-key, comment on the website's native comments feature.

Tokens

Ask claude to implement one of the project sketches on `/problems`. Let us know how it goes.

Podcasts

What podcasts should we go on to discuss this? Get us invited.

I'm easy to find

`quinn@for-all.dev`